Preface from Author

From Static Diagrams to Executable Knowledge: A Personal Journey

For nearly thirty years, I have lived and breathed enterprise IT. My career began long before "knowledge graph" was a common term, working across telecom, automotive, finance, and IT outsourcing consulting. For most of those years, my primary tool for making sense of complexity was static diagramming, mostly with Visio, draw.io, or even office suite like Excel and Powerpoint. It was powerful, but it was also… silent. The diagrams described architecture, but they could not execute it. They could not reason, validate, or adapt.

That was the before.

The shift (2021 – 2022) began when I moved from old toolbox to Archi, the open-source ArchiMate modeling tool. Suddenly, I was not just drawing; I was building a **repository**. I discovered the power of a shared, collaborative model – a single source of truth about an enterprise. But even Archi, as good as it is, had limits. The model was still primarily for human consumption. It was a rich, structured database, but it was not a reasoning engine.

It was around this same time that I discovered the **Essential EAS Open Source (Ontology)** – a glimpse of what happens when an EA repository is built on formal semantics. That was the first crack in the old paradigm, learnt and practiced with its modeling tool - Protégé - the first time I met this word.

The catalyst (2023) was the `Pizza.owl` tutorial. Through creating my first own video series on the topic, I got in touch with Michael DeBellis. More importantly, I got my hands on Protégé. For the first time, I was not just modeling structure; I was formalizing **meaning**. I was defining classes, properties, and rules that a machine could use to infer new knowledge, check for contradictions, and navigate relationships intelligently.

The pizza example was simple, but the potential was explosive. I realized that the future of enterprise architecture was not in prettier diagrams, but in **executable knowledge**. With that learning, back to the Essential EAS tool, I realized how powerful of treating enterprise architecture from semantic perspective, and how flexible the tool provided, e.g. if you need one meaning (triple) that EAS built-in meta-model doesn't exist, just create one new from it's slots and that's now fitting the knowledge of your enterprise reality, more and more.

The synthesis (2023 – 2026) became a period of intense exploration. My YouTube channel (and other video sharing platform, like BiliBili, DouYin, etc..) grew to host not just the Pizza.owl series, but courses on ArchiMate, meta-modeling, Python / C / C++ / Java, PlantUML (diagramming as code), Neo4j (graph databases), and even 3D modeling with OpenSCAD. Each course was a piece of a larger puzzle. programming taught me logic, PlantUML taught me that diagrams could be generated from text (code), Neo4j showed me how to traverse relationships at scale, and OpenSCAD reinforced the power of parametric, declarative design.

All these threads eventually wove themselves into a single, coherent idea: **EKA – Executable Knowledge Architecture** (https://xiaoqi.com). EKA is the formalization of everything I have learned. It is the recognition that an ontology is not a document, but an executable specification. It is the understanding that $EKA = (K, R, \Theta, \Phi, \Gamma)$ – a framework where Knowledge, Reasoning, Triggers, Actions, and Governance become a single, integrated system.

Why this book? This book is not a transcript of my videos. It is the theory behind the practice. It is the result of years of hands-on work, distilled into a structured, pedagogical journey. I wrote it because I saw a gap:

> countless tutorials show you how to click buttons in Protégé, but almost none explain the architectural thinking behind the clicks.

This book aims to bridge that gap, using the familiar `Pizza.owl` as a safe and repeatable domain to teach the foundational principles of EKA.

You will start with pizza. But if you follow the journey, you will be able to model anything: an enterprise architecture, a digital twin, a regulatory framework, or an AI knowledge system.

The goal is not just to teach you Protégé. The goal is to give you a new way to think about knowledge: as something that can be **formalized, reasoned over, governed, and eventually executed.**

Welcome to the journey.

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